Auss Abbood

Cambridge, UK — contact@aussabbood.com — www.aussabbood.com

EDUCATION

University of Cambridge, Cambridge, UK — Oct. 2024 — present
PhD in Computation, Cognition and Language

Osnabrück University, Osnabrück, Germany — Oct. 2016 — May 2019
M. Sc. in Cognitive Science — Grade: 1.1, in a scale of 1 – excellent to 5 – bad
Specialized in artificial intelligence, neuroinformatics, and neuroscience. — With distinction

Goethe University Frankfurt, Frankfurt am Main, Germany — Oct. 2013 — Sep. 2016
B. Sc. in Biological Sciences — Grade: 1.6, in a scale of 1 – excellent to 5 – bad
Specialized in neuroscience.

PUBLICATIONS

- Hicketier, A., Bach, M., Oedi, P., Ullrich, A., **Abbood, A.** (2025). Ensemble-labeling of infectious disease time series to evaluate early warning systems. *Infectious Disease Modelling, 11 (3).*
- **Abbood, A.**, Meng, Z., Collier, N. (2025). *Time* to Revisit Exact Match. *Findings of EMNLP 2025.*
- Bhatia S, [...], **Abbood, A.** [...] et al. (2023). Lessons from COVID-19 for rescalable data collection. *The Lancet Infectious Diseases, 23 (9).*
- **Abbood, A.**, Ullrich A., Denkel, L.A. (2023). Understanding COVID-19 reporting behaviour to support political decision-making: a retrospective cross-sectional study of COVID-19 data reported to WHO. *BMJ Open, 31 (1).*
- **Abbood, A.**, Ghozzi, S. (2023). General Framework for Evaluating Outbreak Prediction, Detection, and Annotation Algorithms. *medRxiv.*
- Oala L, [...], **Abbood, A.** [...] et al. (2021). Machine Learning for Health: Algorithm Auditing & Quality Control. *Journal of Medical Systems, 45 (2021).*
- **Abbood, A.**, Ullrich, A., Busche R., Ghozzi, S. (2020). EventEpi—A natural language processing framework for event-based surveillance. *PLOS Computational Biology, 16 (11).*

EXPERIENCE

Robert Koch Institute — Berlin, Germany
Data Scientist and Project Lead — Oct. 2018 - Sep. 2024

- Built a risk assessment pipeline for COVID-19 by combining tabular and text data, applying self-developed risk metrics, and predicting short-term risk, saving 1 h of manual work daily.
- Co-developed intensivregister.de (ICU surveillance) working on data engineering, visualization, and on-demand data analyses to support ICU management in Germany, serving up to several million visits per day.
- Automated key information extraction and relevance scoring for public health surveillance to potentially save epidemiologists 30 minutes of daily work.
- Built collaborations with partners from the public and private sector from more than a dozen countries and with organisations such as the European Centre for Disease Prevention and Control, Africa Centres for Disease Control and Prevention, and the World Health Organization Hub.
- Led and supervised applied data-science work, including supervision of student assistants and data scientists.

Communintelligence — Münster, Germany
Co-founder and Data Scientist — Oct. 2017 - Jan. 2019

- Data science and deep learning consultation for developers in a public utility company.
- Predicted bus delays with a mean error of only two minutes by analysing multivariate bus traffic time series using random forests.

PROJECTS

Public Health Intelligence Capacity Building and Innovation — Berlin, Germany
Project Link: PHI-CBI — Jan. 2023 - Oct. 2024

- Led technical work to improve user relevance in the text-based surveillance system Epidemic Intelligence from Open Sources (EIOS).
- Conducted user interviews and extensive data analyses to identify bottlenecks in EIOS-based surveillance workflows.
- Evaluated recommender systems, anomaly detection, and text classification approaches to improve the relevance of articles shown in EIOS.

Data and AI-supported early warning system to stabilize the German economy — Berlin, Germany
Project Link: DAKI-FWS — Dec. 2021 - Oct. 2024

- Co-led work packages on early warning from epidemiological data such as Germany's notification system, wastewater surveillance, and self-reported symptom data.
- Implemented a forecast hub with specialized metrics to evaluate AI models for early warning on multimodal data.
- Developed a post-hoc labelling algorithm for COVID-19 time series to evaluate AI-based early-warning models.

Natural Language Processing for Event-based Surveillance with Africa CDC Berlin, Germany
Project Link: NaLaA May 2021 - Jan. 2024

- Led a project applying natural language processing to Twitter data for event-based surveillance.
- Developed a language model in Sesotho for geo-tagging of Tweets.
- Developed an interface integrating a Twitter surveillance component into Africa CDC's event management system.
- Co-organised workshop on event-based surveillance for Southern African RCC on surveillance methods.
- Served as project lead.

The Focus Group on Artificial Intelligence for Health by ITU and WHO
Project Link: FG-AI4H Nov. 2019 - Aug. 2023

- Co-led the Topic Group on Outbreaks and AI test specifications.
- Defined best practices for developing and benchmarking outbreak-detection algorithms.
- Developed a benchmarking setup combining qualitative and quantitative evaluation measures.
- Contributed to best-practice guidance for safe AI use and benchmarking through testing.

CONFERENCE PRESENTATIONS

- Zhang M, **Abbood, A.**, Meng Z., Collier, N. (2025). The Power of Simplicity in LLM-Based Event Forecasting. REALM – Workshop at ACL.
- 35th NeurIPS – Online (2021). Workshop presentation.
- EPIDEMICS 8 – Online (2021). Pre-recorded presentation.
- Applied Machine Learning Days – Lausanne (2020). Poster presentation.
- European Scientific Conference on Applied Infectious Disease Epidemiology – Stockholm (2019). Presentation.
- The 14th Annual Conference of the German Society for Epidemiology – Ulm (2019). Presentation.

GRANTS AND FUNDED PROJECTS

- PHI-CBI: Contributed to successful acquisition of €1.6M for public health intelligence capacity building and innovation, 2023–2024.
- DAKI-FWS: Contributed to successful acquisition of €860K for data- and AI-supported early-warning systems, 2021–2024.
- NaLaA: Led acquisition of €590K for NLP-based event-based surveillance with Africa CDC, 2021–2024.

TEACHING AND SUPERVISION

- Teaching assistant, Li 18: Computational Linguistics, University of Cambridge.
- Lecture on outbreak detection for public health surveillance at the Robert Koch Institute.
- Training of trainers on data science in epidemiology for epidemiologists in Uzbekistan.
- Supervised five data scientists and one student assistant; co-supervised one Master's thesis.
- Co-organised workshop on event-based surveillance methods for Africa CDC Southern Africa RCC.

AWARDS AND SCHOLARSHIPS

- CAM-DTP PhD studentship funded by the ESRC.
- Winner, MOOD EU Horizon 2020 summer school hackathon, for an ensemble ML model predicting tick prevalence from spatio-temporal earth observation, vegetation, and weather data, 2023.
- Rosa-Luxemburg-Foundation Scholarship, 2018.
- Second place, hack4health, for a Google Trends-based early-warning system and chatbot for syndromic surveillance, 2018.
- Winner, MÜNSTERHACK, for AR-based public transport information and ML-based bus-delay prediction, 2017.
- Winner, Hasso-Plattner-Institute Hackathon, for a vaccine-consultation chatbot, 2017.

TECHNICAL SKILLS

- **Programming languages:** Python, R, Java, SQL, Bash
- **Software packages:** Altair, BeautifulSoup, gensim, Flair, Git, Luigi, matplotlib, NLTK, numpy, pandas, plotly, PyTorch, SciPy, Selenium, sklearn, spaCy, Statsmodels, TensorFlow
- **Languages:** German (fluent), English (fluent), Arabic (good)